

Milestone 2: completed and remaining work

A concise status document for customer intelligence, forecasting, backend integration, frontend connection and final evaluation readiness.

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ML workstreams built	Role-scoped forecast APIs	Automated tests passing	Forecast horizons: 7/14/30

What is completed

1	Customer intelligence Customer feature engineering, purchasing-behaviour analysis, engagement scoring and customer segmentation are implemented. K-Means and Hierarchical Clustering were evaluated, and versioned assignments can be imported without duplicates.
2	Revenue forecasting Amazon sales records are cleaned into daily net INR revenue. Seasonal Naive, Prophet, XGBoost and Random Forest are compared with chronological validation and 7/14/30-day predictions are generated.
3	Product-demand forecasting The train and evaluation Parquet datasets are used for lag, calendar, promotion, stock and weather features. The real full-data run selected XGBoost with R2 0.918 across 250 high-volume store/product series.
4	Database and repeatable imports Forecast model runs, per-series metrics, predictions and import jobs are stored through Alembic migrations. Re-importing the same model version updates changed records and creates no duplicates.
5	Four-role backend access Business Owner receives business revenue; Store Manager receives assigned-store demand; Sales Executive receives a personal-only scope; Administrator receives MFA-protected model and job monitoring.
6	Verification Seventeen backend tests and nine preprocessing tests pass. Clean migrations, repeat imports, role restrictions, Administrator MFA and all four live forecast endpoints were verified.

Remaining work

Complete these tasks in order. The forecasting backend is ready; the main critical path is frontend connection and final integration validation.

Priority	Work area	What must be completed	Done when
P0	Forecast frontend	Build/connect the Owner, Manager, Sales Executive and Admin forecast pages. Replace every demo KPI, chart, table and model value with the role-specific API response.	All four pages load real database-backed values.
P0	Filters and UI states	Connect 7/14/30-day horizon, category, store and product controls. Add loading, empty, unauthorized and API-error states. Add CSV export if it remains in the approved design.	Each control changes the API request and visible result.
P1	Personal forecast data	Prepare seller-attributed history and train/import one genuinely personal model scope per Sales Executive. The current demo seller is authorized correctly but uses the shared Amazon forecast source.	Two sellers receive different forecasts from their own history.
P1	Product/inventory mapping	Map source store and product identifiers to MarketMind Store, Product and Inventory records. Unmapped series currently cannot calculate a reliable stock-risk level.	Forecasted products show mapped names, stock and risk.
P1	Admin monitoring UI	Connect model registry, selected metrics, version, training time and import-job status to the monitoring endpoint. Mark scheduling and live latency as future work unless approved for this milestone.	No monitoring card uses fixed demonstration text.
P1	Revenue quality	Do not display fake accuracy. The current short Amazon date window produces negative aggregate R2. Obtain longer history or present the result as an experimental Milestone 2 forecast with its real MAE, RMSE and R2.	UI and documentation state the real limitation.
P0	End-to-end testing	Run clean migrations, seed four roles, import segmentation and forecasts twice, verify RBAC/MFA, build the frontend and test every forecast page and filter.	Clean-install checklist and all automated checks pass.
P2	Documentation and Git	Add final screenshots, API examples, model metrics, known limitations and demo steps. Review the exact diff, then commit to Garvitk001 and integrate only after approval.	Evaluation pack and reviewed branch are ready.

Execution order and completion checks

The sequence below keeps the team from polishing UI against unstable data or repeating integration work.

1**Frontend API connection**

Connect the four forecast pages and all approved controls to the existing FastAPI contract.

2**Identifier and seller mapping**

Finish product/store mapping and confirm the source of each seller-specific training series.

3**Role-by-role review**

Verify permitted data scope, forbidden routes, Administrator MFA and visible navigation for each login.

4**Clean integration run**

Migrate a clean database, seed, import twice, run model/backend/frontend tests and save evidence.

5**Documentation and branch review**

Update screenshots and results, review the exact diff, then commit and integrate after team approval.

Final Milestone 2 acceptance checklist

- Customer segments and engagement use reproducible processed data and versioned model results.
- Revenue, personal and product-demand pages contain no hard-coded forecast values.
- Every forecast shows its model version, generated time, horizon and real evaluation metrics.
- Owner, Manager, Sales Executive and Administrator receive only their allowed data scope.
- Product demand is connected to inventory wherever a reviewed identifier mapping exists.
- A clean database setup and a second import create no duplicate model or prediction records.
- Backend tests, preprocessing tests and the frontend production build pass.
- Known data limitations are visible in the evaluation notes and are not presented as accuracy claims.

Immediate next action

Connect the four forecasting frontend designs to the live APIs, beginning with the Business Owner revenue page, then Sales Executive personal forecast, Store Manager demand forecast and Administrator monitoring.